

HOWARD PAGE

Burlingame, California 94010 | (650) 703-8733 | howard.page@yahoo.com | www.linkedin.com/in/howard-page/

SUMMARY

A highly skilled software engineer with demonstrated expertise in software development, consulting, requirements gathering, system specification, architecture design, end-user documentation, programming, problem solving, project management, and strategic planning. Established capabilities in communication, organization, leadership, teamwork, and collaboration. Excels at prioritizing multiple competing tasks in dynamic environments. Driven to achieve results and exceed expectations to ensure the long-term success of an enterprise.

TECHNICAL SKILLS

Problem Solving | C | C++ | Java | JavaScript | Lua | Python | Linux | VxWorks | XML | JSON | make | Gradle | Maven | GDB | Tomcat | MySQL | MongoDB | Azure | AWS | DRM | CloudFront | HTTP | RTSP | SCTE-35 | ESAM | PKI | VOD | OAuth2 | GNU toolchain | Perforce | Git | Bugzilla | Jira | Jenkins | Docker

KEY ACCOMPLISHMENTS

MPEG, VOD, Interactive Television, Set Top Box / Home Gateway

- Developed:
 - Client voice control subsystem implemented via both Google Voice (Dialogflow) and AWS Alexa (Alexa Skills) (Java, Tomcat, JSON, Gradle, JavaScript, Lua, Azure, Ngnix).
 - Google Widevine DRM server (using Widevine API) and proxy server (from scratch) (Java, Tomcat, Gradle, JSON, AWS, Docker).
 - Multiple enhancements to Vudu Java / Tomcat backend server – port of C++ DRM license server, AWS Cloudfront CDN, AVOD infrastructure, feature tests (C++, Java, Tomcat, Gradle, JSON, AWS, Docker, MySQL).
 - SCTE-35 / ESAM Placement Opportunity Information Service server (Java, Spring, Tomcat, MongoDB, Maven).
 - Video multimedia gateway message processing system (C++, STL, state machine, XML, Git, JSON).
 - ATM attached VOD / switched digital video set top box / home gateway and CPE device (Linux, VxWorks, Java / JNI, Servlet, C, C++, HTTP, RTSP, XML, MPEG-2, VOD, device driver, sockets, bash).
 - TV commercial ad insertion system (Linux, C, C++, device driver).
 - VOD / Web image synchronization platform (Linux, Win98, C, Visual Basic, sockets).
 - VOD set top box user interface system (Solaris, OS/9, C, PERL, ORACLE, MySQL, Postgres).
 - MPEG-2 video encoding station (Solaris, C, ksh).
 - Interactive TV transaction server (Solaris, C, C++, PKI / DES, Tuxedo, sockets).
 - Video Multiprocessing Gateway (C++, Python, Java).
 - Linux MPEG-2 decoder device drivers (Linux, C).
- Digital Video Security System
 - Design, development, and integration of Linux-based MPEG-2 security station and backend (C, C++ / STL, GTK, Qt, Postgres, XML).
- PDA / Pen-Based Computers
 - Developed Google Cast work-alike remote-control for Vudu clients on Android platform (Java, Android).
 - Implemented PenPoint AI neural network handwriting recognizer engine (C++, PenPoint).
 - Developed PenPoint handwriting AI recognition engine testing platform and enhanced telecommunication service application (C, PenPoint).
- Software Development Experience
 - Developed Bluetooth connected embedded Linux power monitoring device.
 - Developed Linux / IBM VM system administration tools (C++ / STL / BOOST, CORBA, VM, Postgres).
 - Developed MPEG-2 / DSM-CC ATM Level 1 Gateway server (Solaris, C, SNMP, MPEG-2).
 - Developed network credit authorization server administrator and configuration interface (QNX, C, LISP).
 - Software development methodology skills and experience, including the development of system requirements, specifications, architecture, and end-user system documentation.
 - Real-time data acquisition and control software development.
 - Experience with many tools, including GNU toolchain, Perforce, Git, Bugzilla, Jira, Jenkins, AWS, Docker, etc.

EXPERIENCE

VUDU / WALMART / FANDANGO / NBCU / COMCAST, San Jose, California

Senior Software Engineer, 2016-2025

- Developed text-to-speech subsystem.
- Developed Android-based remote control for Vudu platform.
- Developed voice control for Vudu platform, using Google Dialogflow / Alexa Skills.
- Designed and implemented Google Widevine server and proxy server. Moving servers in-house saved the company greater than \$100K / year.
- Supported and enhanced Vudu backend system, Vudu platform on Sony PS3 / PS4.

RGB NETWORKS / IMAGINE COMMUNICATIONS, Sunnyvale, California

Senior Software Engineer, 2013-2016

- Architected and developed SCTE-35 messaging feature for communication feature for video multiprocessing gateway product.
- Produced and developed SCTE-35-based placement opportunity information service (POIS) server.

ADDITIONAL EXPERIENCE

SENTIENT ENERGY, Burlingame, California, **Senior Software Engineer**, 2012-2013. Served as software architect / developer for next-generation embedded Linux-based power line monitor.

SEACHANGE INTERNATIONAL (LIBERATE DIVISION), San Mateo, California, **Senior Software Engineer**, 2006-2012. Enhanced WebKit-based DTV browser, using multiple technologies including SIP, UPnP / DLNA, and MHEG application engines.

SOFTIER, INC., Sunnyvale, California, **Senior Member of Technical Staff**, 2005-2006. Integrated DLNA / UPnP video / photo / music player into TI DM642 DSP-based platform.

SEASCAPE COMMUNICATIONS, Menlo Park, California, **Software Consultant**, 2003-2005. Fixed / enhanced player for video / audio on demand system. Developed enhanced player client. Developed process control software.

LINUXCARE, San Francisco, California, **Senior Software Engineer**, 2003. Developed system administration tools for Linux / z/VM.

FUTURETEL DIGITAL IMAGING, Cupertino, California, **Software Consultant / Technical Lead**, 2002. Led integration of Linux-based MPEG-2 digital video security system.

SPRINT ADVANCED TECHNOLOGY LABORATORIES, Burlingame, California, **Software Consultant / Technical Lead**, 1996-2002. Principal software designer / developer for Sprint ATL broadband FTTH VOD / switched digital video / telephony set top box / home gateway. Developed television and download / scheduling / insertion system. Developed web-based VOD controller. Developed VOD / web browser media synchronization system. Developed network intelligent agent demonstration.

SUN MICROSYSTEMS, Mountain View, California, **Software Consultant**, 1995-1996. Developed MPEG-2 trick play stream encoding process. Developed OpenTV UI user interface application. Designed and developed interactive TV transaction server.

VISA INTERNATIONAL, San Mateo, California, **Software Consultant**, 1994-1995. Designed and implemented the configuration subsystem for VisaNet Access Point network device. Developed network attached interface.

GO CORPORATION, Foster City, California, **Software Consultant**, 1993. Developed PenPoint 3.0 telecommunications enhanced service and handwriting recognizer testing applications.

AT&T BELL LABORATORIES, Holmdel, New Jersey, **Software Consultant**, 1991-1993. Ported AI neural network-based handwriting recognizer to PenPoint. Developed PenPoint handwriting recognition test platform. Productized AI neural network based OCR system.

IBM T.J. WATSON RESEARCH CENTER, Hawthorne, New York, **Software Consultant**, 1990-1991. Participated in the effort to develop an AIX / X Windows pen computer system.

EDUCATION

STEVENS INSTITUTE OF TECHNOLOGY, Hoboken, New Jersey
Masters (M.S.), Computer Science

UNIVERSITY OF COLORADO BOULDER, Boulder, Colorado
Bachelors (B.A.), Computer Science

CERTIFICATION

Google Widevine Certified

PATENTS

US5,784,504: Disambiguating input strokes of a stylus-based input devices for gesture or character recognition
US5,825,884: EP0847649B1: Method and apparatus for operating a transactional server in a proprietary database environment
US6,970,535: Wireless messaging system to multiple recipients
US8,418,195: Advertising insertion for a video-on-demand system
EP0895694B1: System and method for creating trick play video streams from a compressed normal play video bit stream